

Exercise: SpaceX Launches — Python CLI with argparse

Overview Build a command-line tool (spacex.py) that fetches SpaceX launch data from the public API and produces useful summaries. The tool should be user-friendly, resilient to network issues, and support a verbosity mode for debugging.

API <https://api.spacexdata.com/v4/launches>

Goals

- Fetch and cache SpaceX launches JSON.
- Filter launches for the year 2022
- Print different reports based on an `--action` argument.
- Support `--verbose` logging and basic resilience (timeouts, retry).

Usage Examples

- `python spacex.py --action report`
- `python spacex.py --action payloads -v`
- `python spacex.py --action launchpads --refresh --cache .cache/launches.json`

CLI Arguments

- `--action {report, payloads, launchpads}` (required)
- `--verbose` or `-v` (optional): increases logging detail
- `--refresh` (optional): ignores cache and refetches from API
- `--cache PATH` (optional): path to cache file (default: `launches.json`)

Actions and Outputs

- `report`:
 - Year 2022 summary:
 - Total launches
 - Successful launches
 - Failed launches

- Success ratio (%) where denominator excludes launches with success == None
- Example output:
 - Total: 6 | Successful: 5 | Failed: 1 | Success ratio: 83%
- payloads:
 - Average payload count per launch
 - Treat missing payloads as zero.
 - Example output:
 - Average payloads per launch:1.67
- launchpads:
 - Count of launches per launchpad ID.
 - Sort counts descending; use “unknown” if launchpad is missing.
 - Example output:
 - 5e9e4502f509094188566f88 — 4
 - 5e9e4501f509094ba4566f84 — 2
 - unknown — 1

Behavior Details

- Data source:
 - HTTP GET to <https://api.spacexdata.com/v4/launches> returning a JSON list.
- Caching:
 - Save the raw JSON to a local file after a successful fetch (default launches.json).
 - Use the cache unless --refresh is provided.
- Filtering:
 - Consider only launches with date_utc within the year 2022
 - Skip entries with invalid or missing dates.
- Robustness:
 - HTTP timeout: 15 seconds; retry once on timeout.

- On non-200 responses: print a clear error and exit with a non-zero code.
- Handle missing fields safely